📘 **AI/ML Internship – Day 43 Notes & Tasks**

🟢 **Module 7: Support Vector Machine (SVM) Classification Algorithm**

📌 Part 1: Introduction to SVM

🔹 What is SVM?

👉 SVM (Support Vector Machine) is a Supervised Machine Learning algorithm used mainly for:

- Classification
- Regression

👉 It is most commonly used for classification tasks.

🔹 Full Form

SVM = Support Vector Machine

📌 Part 2: Real-Life Example

Imagine two groups of students:

Group A = Fail

Group B = Pass

SVM tries to draw the **best boundary (line)** between the two groups.

Example:

Fail Students | Pass Students

```
● ● ●        |        ▲ ▲ ▲
● ●         |        ▲ ▲
```

The vertical line in the middle is called the **Decision Boundary**.

📌 Part 3: Theory Answers (IMPORTANT)

🔹 **1. What is SVM?**

👉 SVM is a supervised learning algorithm used to classify data by finding the best decision boundary.

🔹 **2. What is a Decision Boundary?**

👉 A decision boundary is a line or plane that separates different classes.

🔹 **3. What are Support Vectors?**

👉 Support vectors are the data points closest to the decision boundary.

These points help SVM create the best separation.

🔹 **4. Is SVM Supervised or Unsupervised?**

👉 SVM is a Supervised Learning algorithm.

🔹 **5. Why is SVM Popular?**

👉 Because:

- High accuracy
- Works well with small datasets
- Effective in classification tasks

📌 **Part 4: Important Concepts**

🔹 **Decision Boundary**

Class A | Class B

● ● ● | ▲ ▲ ▲
● ● | ▲ ▲

The line separating both classes is called the decision boundary.

🔹 **Support Vectors**

● ● ● ● | ▲ ▲ ▲ ▲

The closest points near the boundary are support vectors.

🔹 **Margin**

👉 Margin is the distance between support vectors and the decision boundary.

SVM tries to maximize this margin.

📌 **Part 5: Visualization**

Class A

● ●

--------------------- ← Decision Boundary

▲ ▲

Class B

📌 **Part 6: Example Dataset**

Study Hours	Result
1	Fail
2	Fail
3	Fail
5	Pass
6	Pass
7	Pass

📌 **Part 7: Import Libraries**

```
import pandas as pd

from sklearn.svm import SVC
```

📌 Part 8: Create Dataset

```
data = {
  "Hours":[1,2,3,5,6,7],
  "Result":[0,0,0,1,1,1]
}

df = pd.DataFrame(data)

print(df)
```

Meaning

Value	Meaning
0	Fail
1	Pass

📌 Part 9: Features and Labels

```
X = df[["Hours"]]

y = df["Result"]
```

📌 Part 10: Create SVM Model

```
model = SVC()
```

📌 Part 11: Train the Model

```
model.fit(X, y)
```

📌 Part 12: Make Prediction

```
prediction = model.predict([[4]])

print(prediction)
```

Possible Output:

[1]

Meaning:

👉 PASS

📌 **Part 13: Predict Multiple Values**

```
prediction = model.predict([[2],[4],[8]])

print(prediction)
```

Possible Output:

[0 1 1]

📌 **Part 14: Complete Program**

```
import pandas as pd
from sklearn.svm import SVC

data = {
   "Hours":[1,2,3,5,6,7],
   "Result":[0,0,0,1,1,1]
}

df = pd.DataFrame(data)

X = df[["Hours"]]
y = df["Result"]

model = SVC()

model.fit(X, y)
```

```python
prediction = model.predict([[5]])

print("Prediction:", prediction)
```

📌 Part 15: Advantages of SVM

✅ High Accuracy

✅ Works well with small datasets

✅ Effective for classification

✅ Handles high-dimensional data

✅ Good generalization

📌 Part 16: Disadvantages of SVM

❌ Slower on large datasets

❌ Difficult to understand mathematically

❌ Requires parameter tuning

📌 Part 17: Real-World Applications

📧 Spam Detection

Classify emails as:

- Spam
- Not Spam

😊 Sentiment Analysis

Classify reviews as:

- Positive
- Negative

🏥 Healthcare

Disease Classification

👤 Face Recognition

Identify persons from images.

💳 Fraud Detection

Detect fraudulent transactions.

📌 Part 18: SVM Workflow

Dataset
↓
Features (X)
↓
Find Best Boundary
↓
Train Model
↓
Prediction

📌 Part 19: SVM vs Logistic Regression

SVM	Logistic Regression
Uses decision boundary	Uses probability
High accuracy	Simpler
Good for complex data	Easy to interpret
Works well in high dimensions	Faster training

📌 Part 20: SVM vs KNN

SVM	KNN
Creates boundary	Uses neighbors
Faster prediction	Slower prediction
Good for high-dimensional data	Good for small datasets
Model-based	Instance-based

📌 Part 21: Why SVM is Important?

👉 SVM is widely used in:

- Image Recognition
- Text Classification
- Medical Diagnosis
- Financial Prediction

Because it often provides excellent classification performance.

🎯 Tasks for Day 43

🧠 Task 1: Theory

1. What is SVM?
2. What is a Decision Boundary?
3. What are Support Vectors?
4. What is Margin?
5. Advantages of SVM?

💻 Task 2: Dataset Creation

1. Student Result Dataset
2. Disease Dataset
3. Spam Detection Dataset

💻 Task 3: Model Building

1. Create SVM model
2. Train model
3. Make predictions

💻 Task 4: Prediction Practice

Predict:

1. Student Result
2. Disease Status
3. Spam Detection

💻 Task 5: Compare Algorithms

Compare:

1. Logistic Regression
2. KNN
3. Decision Tree
4. Random Forest
5. SVM

💻 Task 6: Visualization Activity

Draw:

- Decision Boundary
- Support Vectors
- Margin

using a notebook or paper.